Evading next-gen AV using A.I.

https://github.com/endgameinc/gym-malware

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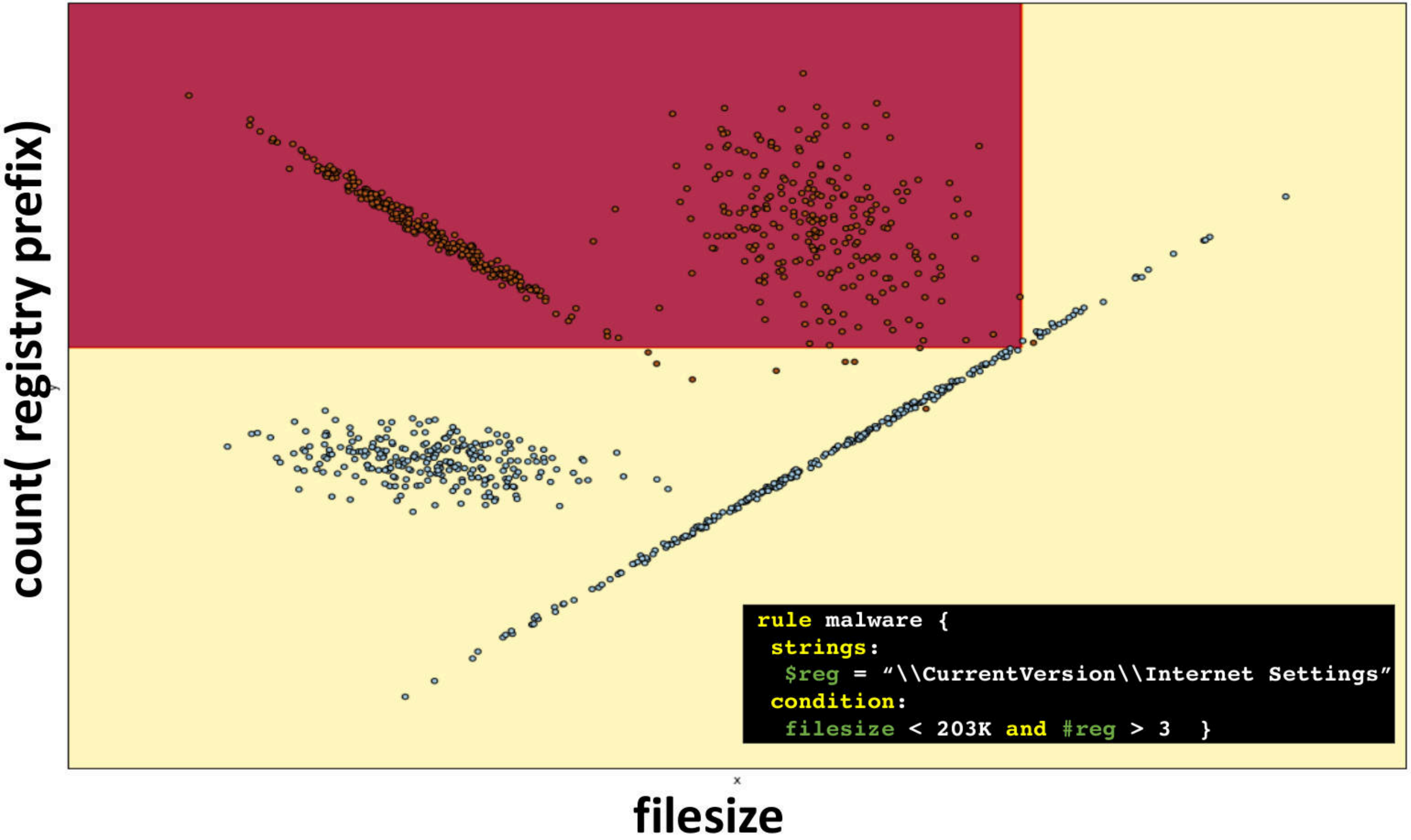
Rules for static analysis

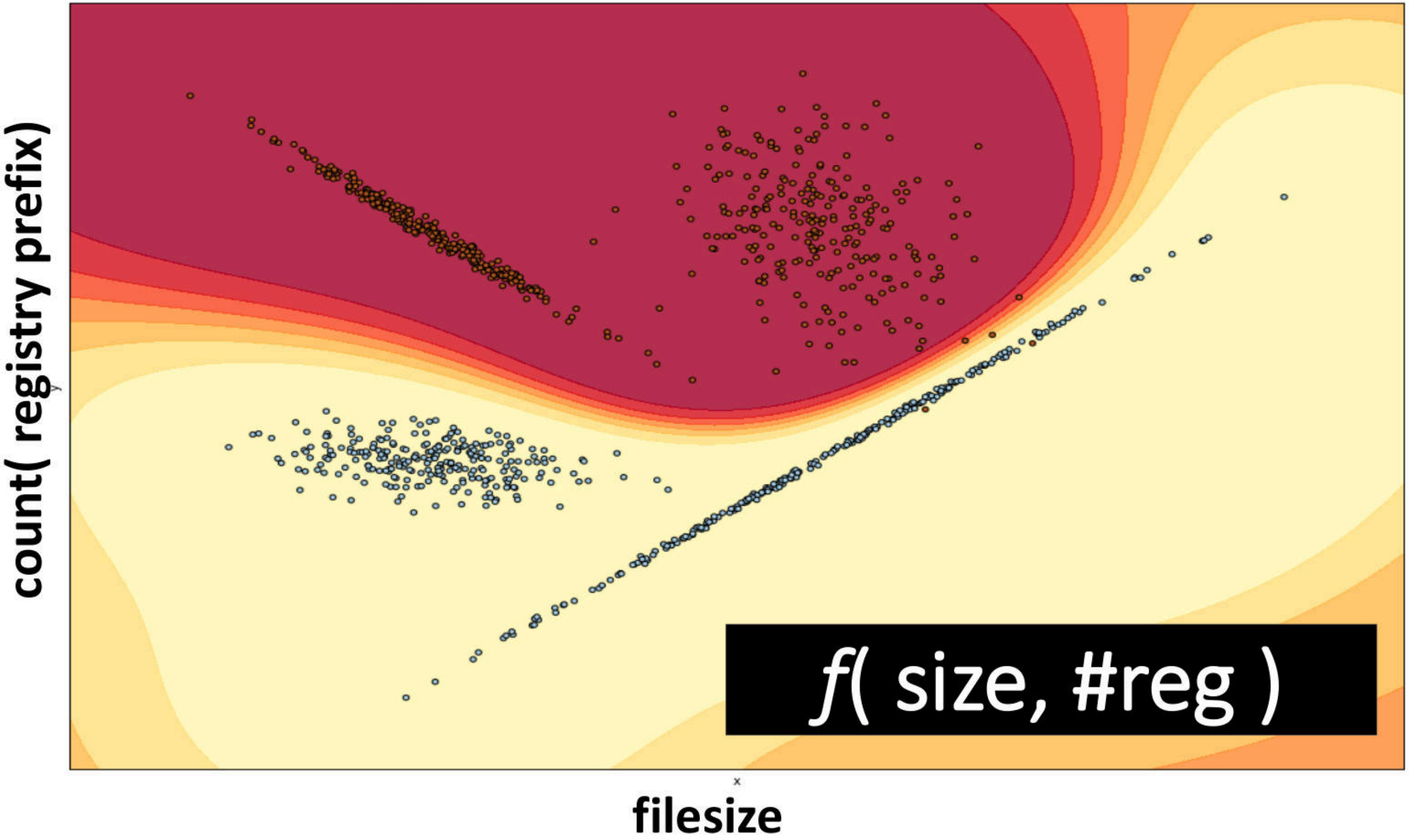


Next-gen AV using machine learning

Automated

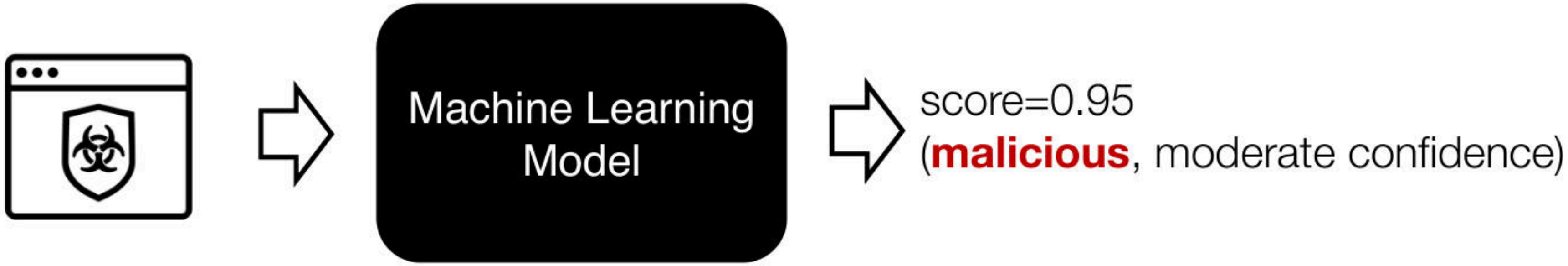
Sophisticated relationships

Generalizes

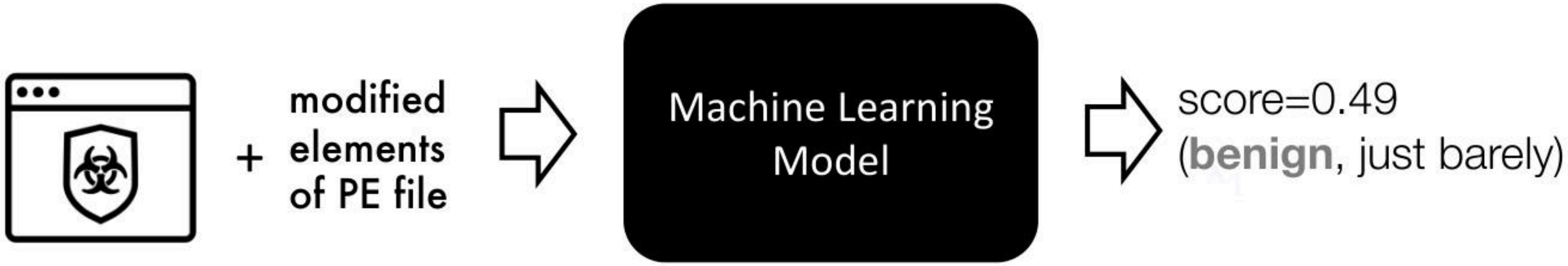


Can We Break Machine Learning?

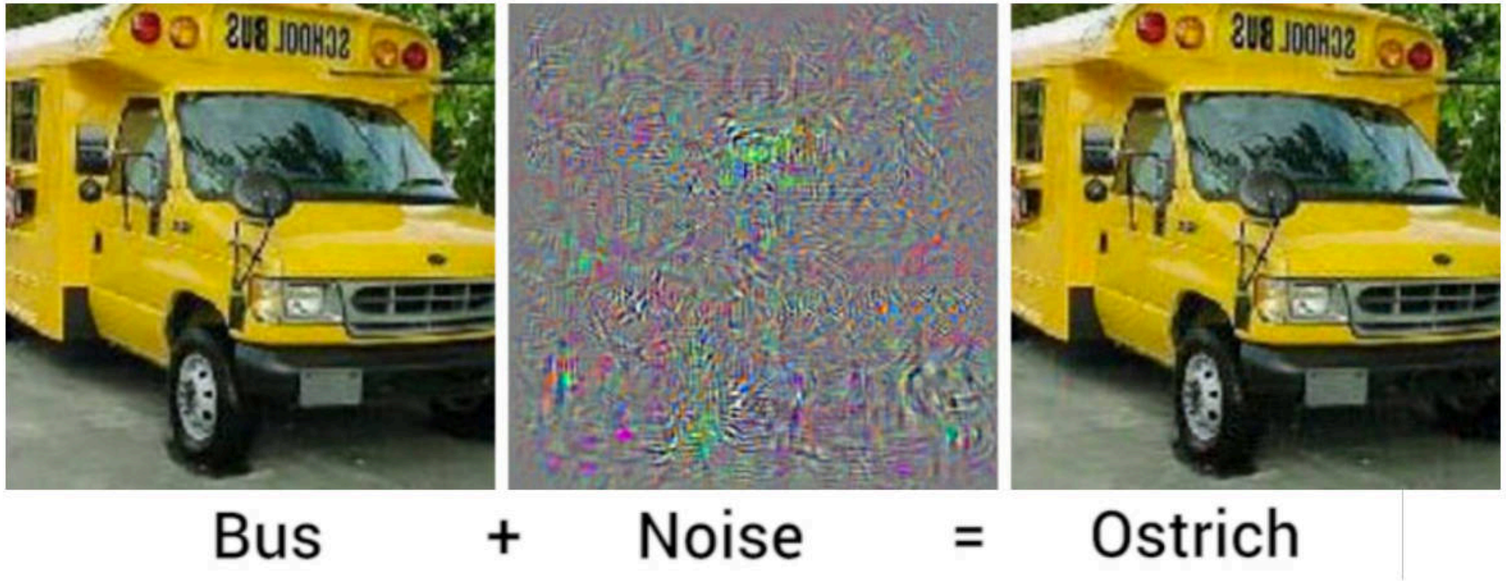
- Static machine learning model trained on millions of samples



- Simple structural changes that don't change behavior
  - upx_unpack
  - '.text' -> '.foo' (remains valid entry point)
  - create '.text' and populate with '.text from calc.exe'



Yes! For deep learning and images...

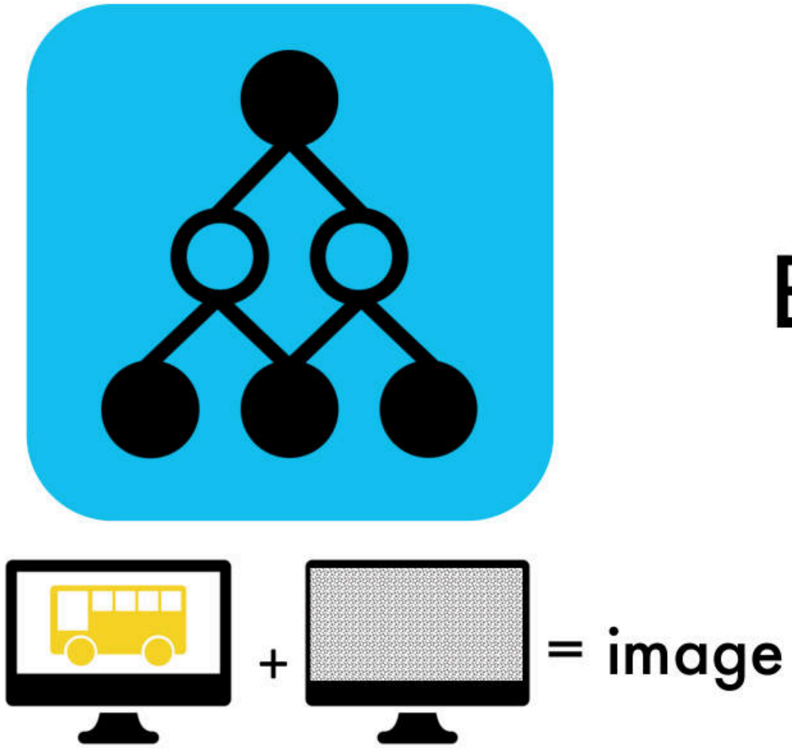


- Machine learning models have blind spots / hallucinate (modeling error)
- Depending on model and level of access, they can be straightforward to exploit
- Adversarial examples can generalize **across** models / model types (Goodfellow 2015)
  - blind spots in MY model may also be blind spots in YOUR model

image credit: http://www.popsci.com/byzantine-science-deceiving-artificial-intelligence

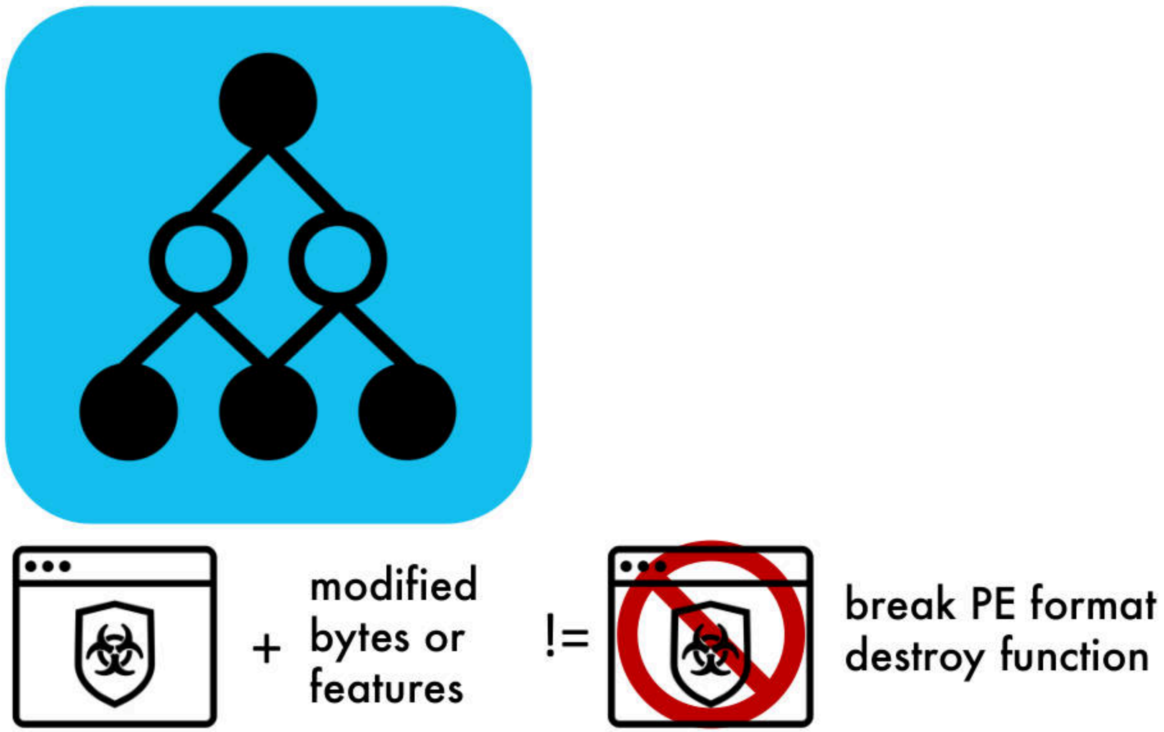
What about for PE malware?

Bus (99%), Ostrich (1%)



BUT...

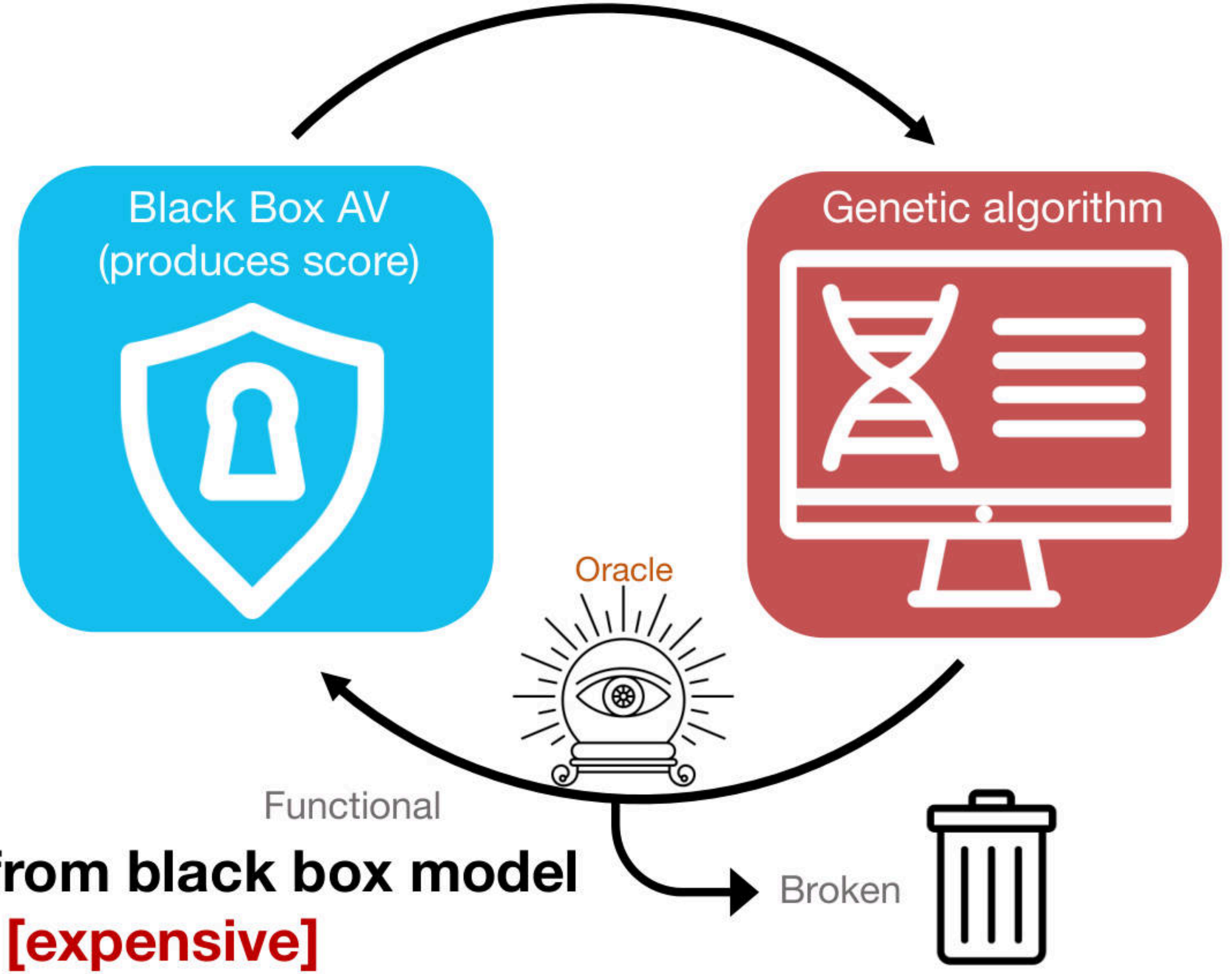
Malware (90%), Benign (10%)



Attacker requires full knowledge of *deep learning* model

Generated sample may not be valid PE file

Related Work: PDF attack / report score



Requires score from black box model
Oracle/sandbox [expensive]

EvadeML [for PDF malware]
(Xu, Qi, Evans, 2016)

Summary of Previous Works

Gradient-based attacks: perturbation or GAN

- Attacker requires **full knowledge** of model structure and weights
- Generated sample may not be valid PE file

Genetic Algorithms

- Attacker requires **score** from black box model
- Requires oracle/sandbox [expensive] to ensure that functionality is preserved

Goal: Design a machine learning agent that

- bypass **black-box** machine learning using
- **format-** and **function-preserving** mutations
- **hardest attack to pull off; difficult to learn**

Reinforcement Learning!

Atari Breakout

Nolan Bushnell, Steve Wozniak, Steve Bristow

Inspired by Atari Pong

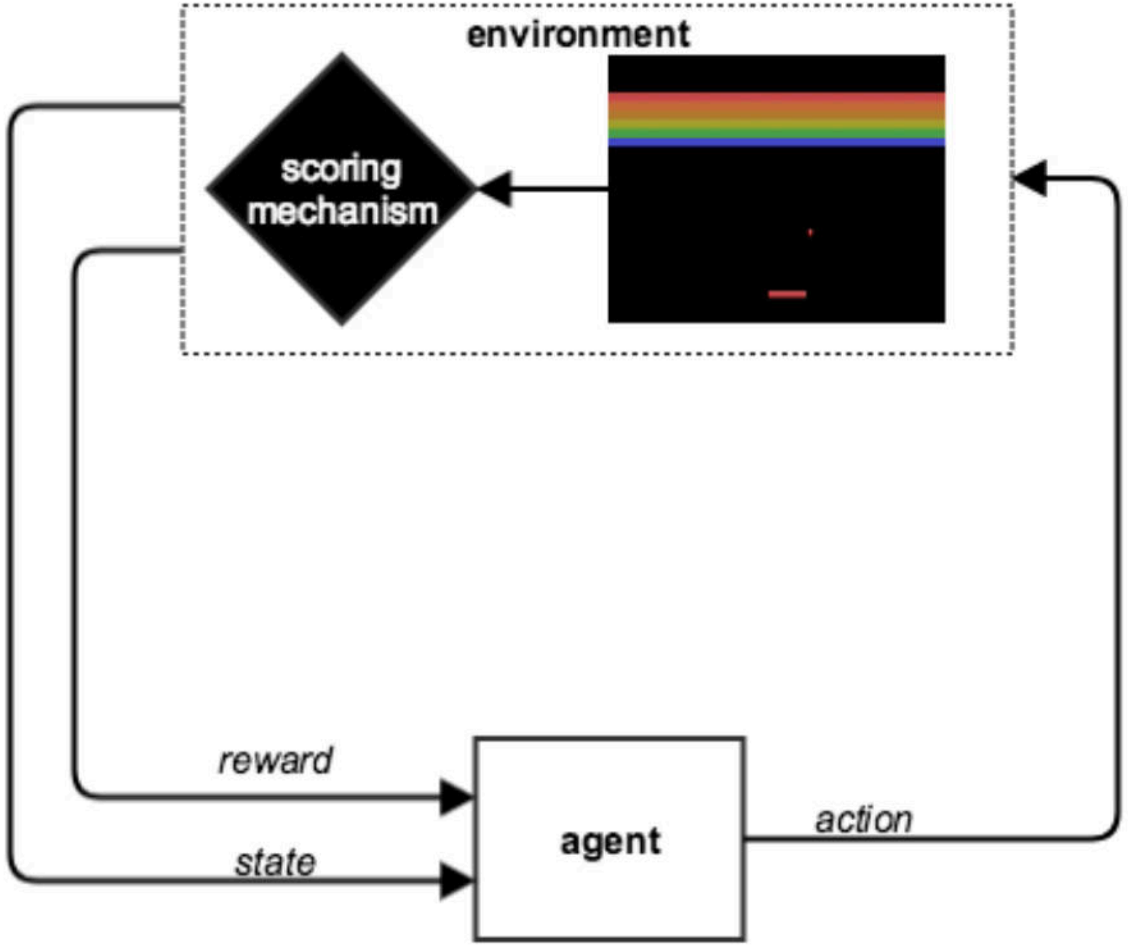
"A lot of features of the Apple II went in because I had designed Breakout for Atari"

(The Woz)

Game

- Bouncing ball + rows of bricks
- Manipulate paddle (left, right)
- Reward for eliminating each brick

Atari Breakout: an AI

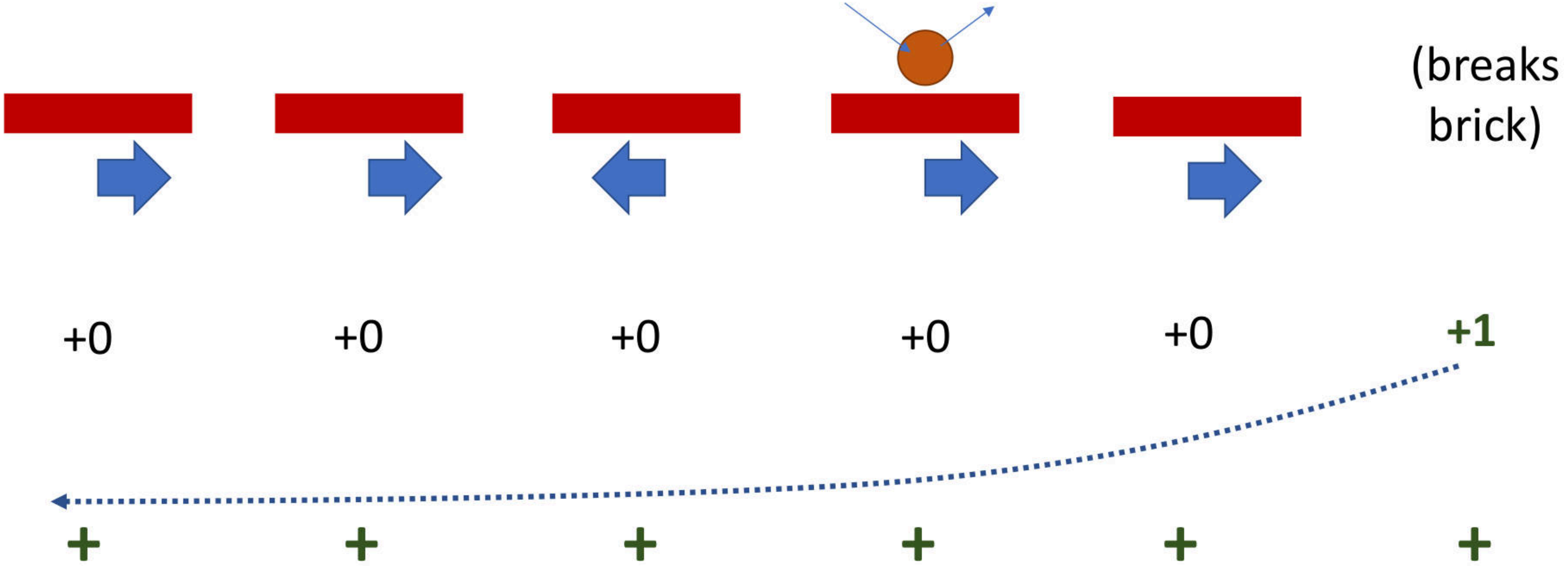


- **Environment**
  - Bouncing ball + rows of bricks
  - Manipulate paddle (*left, right, nothing*)
  - Reward for eliminating each brick
- **Agent**
  - Input: **environment state** (*pixels*)
  - Output: **action** (*left, right*) via **policy**
  - Feedback: delayed **reward** (*score*)
- Agent learns through 1000s of games:

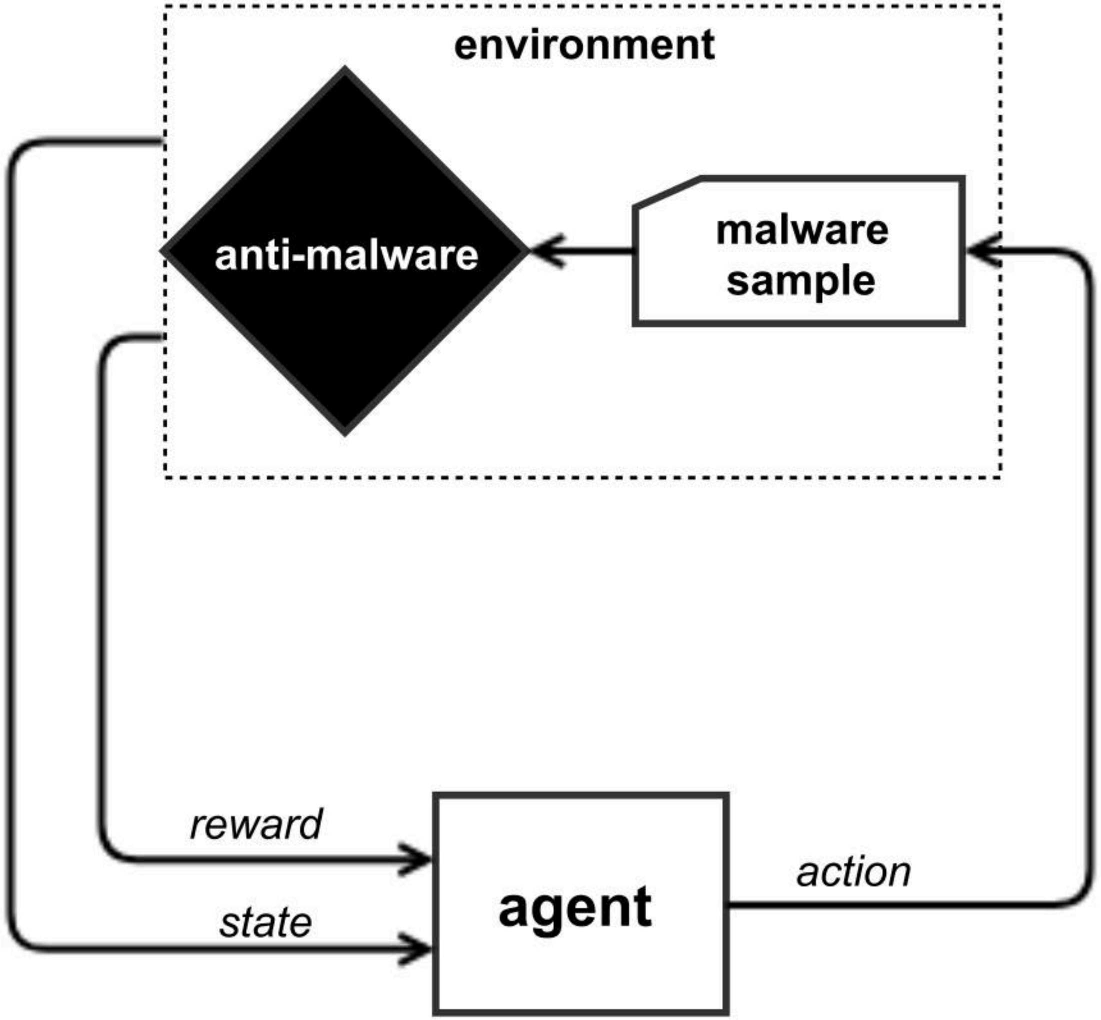
  what action is most useful given a screenshot of the Atari gameplay?

  https://gym.openai.com/envs/Breakout-v0

Learning: rewards and credit assignment



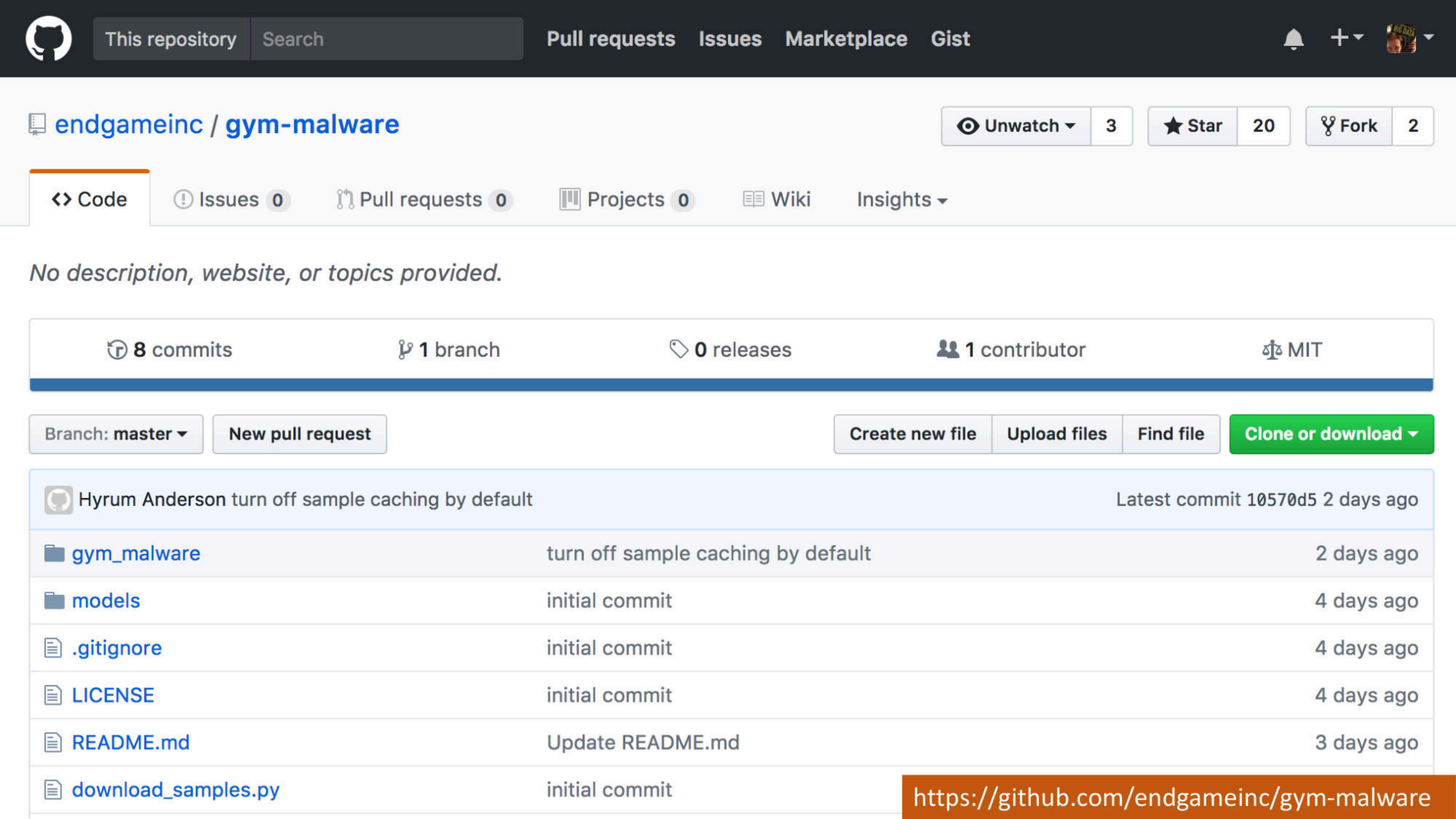
Anti-malware evasion: an AI



- **Environment**
  - A malware sample (*Windows PE*)
  - Buffet of malware mutations
    - *preserve format & functionality*
  - Reward from static malware classifier

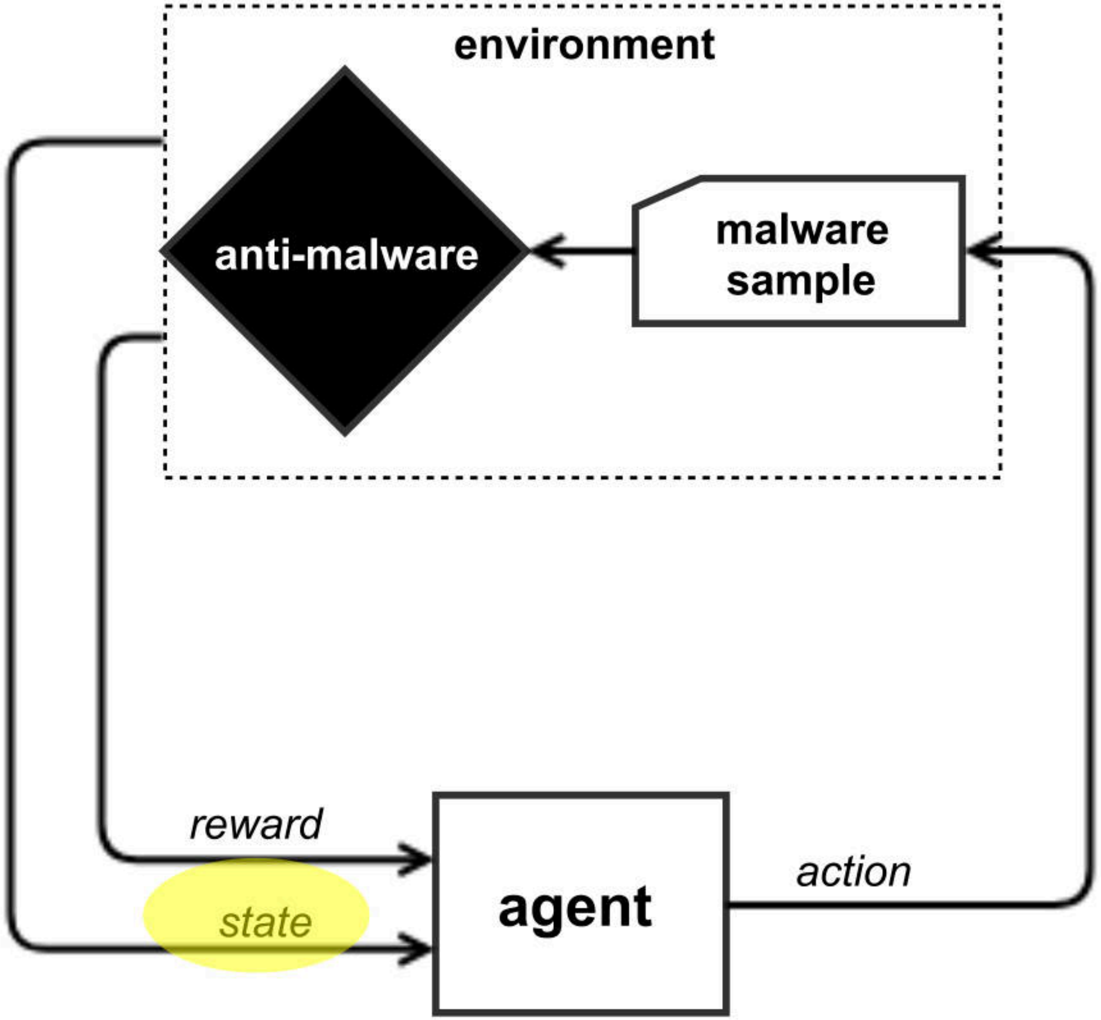
- **Agent**
  - Input: **environment state** (*malware bytes*)
  - Output: **action** (*stochastic*)
  - Feedback: **reward** (*AV reports benign*)

https://github.com/endgameinc/gym-malware



https://github.com/endgameinc/gym-malware

The Agent's State Observation

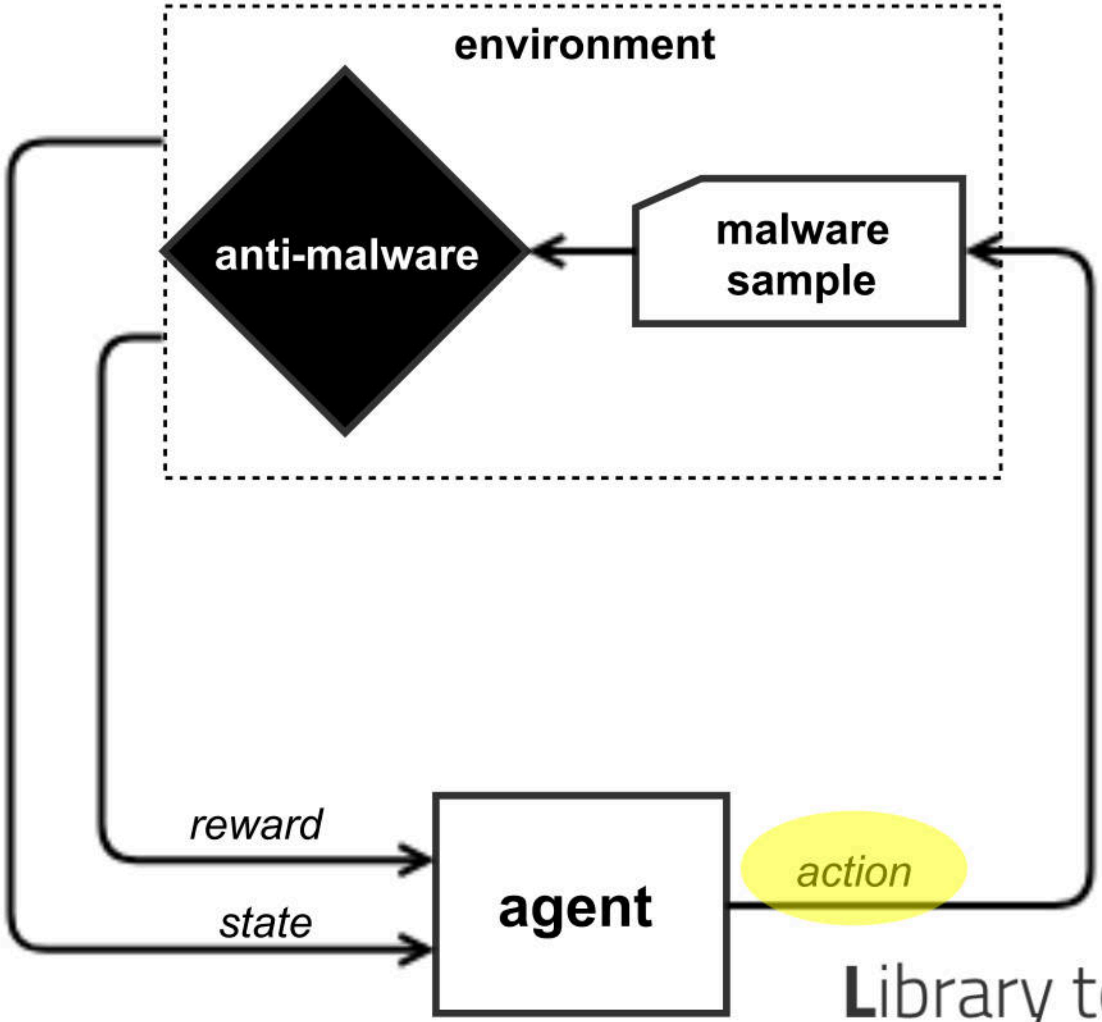


Features

- Static Windows PE file features compressed to 2350 dimensions
  - General file information (size)
  - Header info
  - Section characteristics
  - Imported/exported functions
  - Strings
  - File byte and entropy histograms

- Feed a neural network to choose the best action for the given "state"

The Agent's Manipulation Arsenal



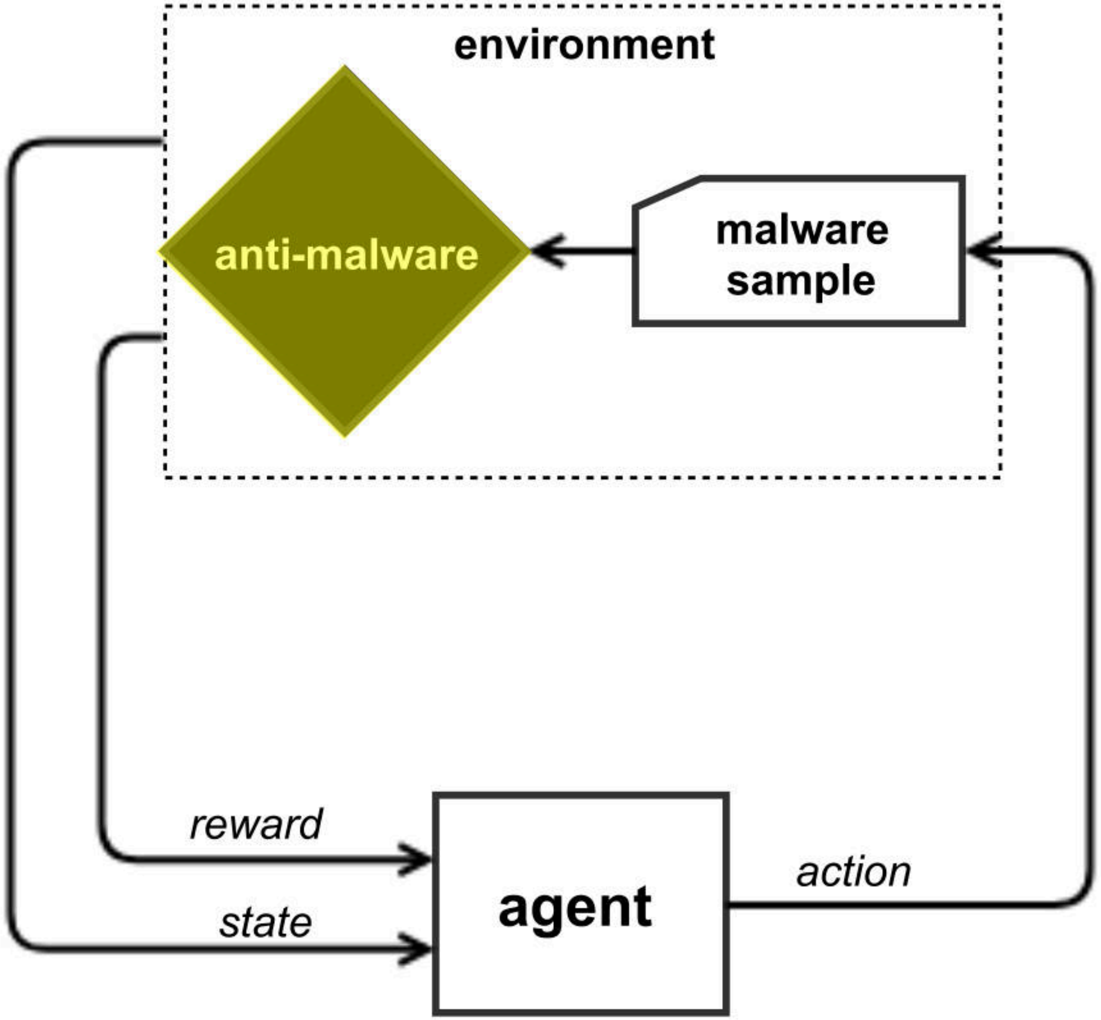
Functionality-preserving mutations:

- **Create**
  - New Entry Point (w/ trampoline)
  - New Sections
- **Add**
  - Random Imports
  - Random bytes to PE overlay
  - Bytes to end of section
- **Modify**
  - Random sections to common name
  - (break) signature
  - Debug info
  - UPX pack / unpack
  - Header checksum
  - Signature

Quarkslab

Library to Instrument Executable Formats

The Machine Learning Model



Static PE malware classifier

- gradient boosted decision tree (for which one *can't directly do gradient-based attack)*
- need not be known to the attacker

Machine learning malware model for demo purposes only. Resemblance to Endgame or other vendor models is incidental.

Ready, Fight!

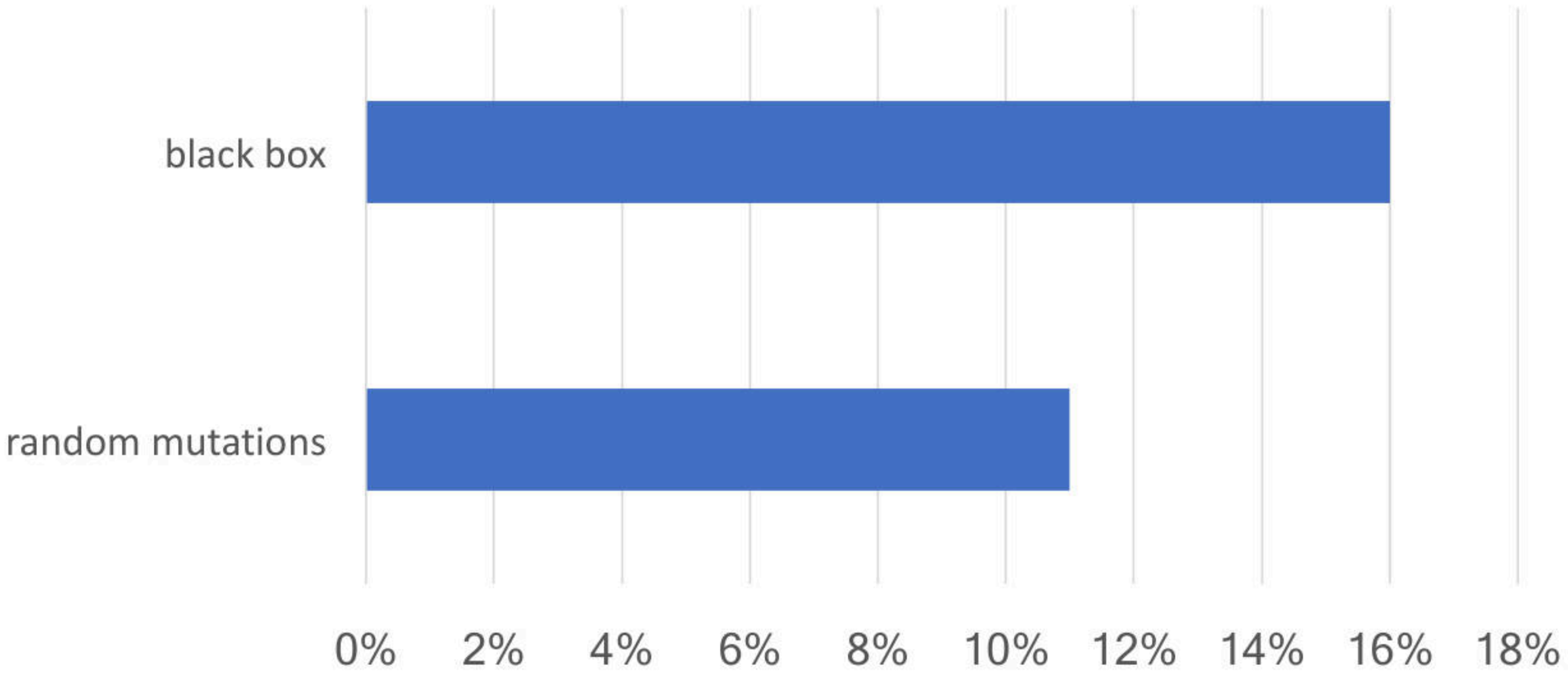
```
:~/src/external/gym-malware$ python train_agent_chainer.py
 new sha256: 023a54b6b8b405f7481d96dfd145b0bbe4b8673da73943a460e3635fd2dacb0a
upx_unpack
 new hash: 023a54b6b8b405f7481d96dfd145b0bbe4b8673da73943a460e3635fd2dacb0a
 score=0.9920191235238174 (hidden), label=1.0
create_new_entry
 new hash: 0dbd26e398fb21e58e7a555aed4556c4683e630f48bb21370ce461d2bfbc31b5
 score=0.9922864489998511 (hidden), label=1.0
upx_unpack
 new hash: 0dbd26e398fb21e58e7a555aed4556c4683e630f48bb21370ce461d2bfbc31b5
 score=0.9922864489998511 (hidden), label=1.0
upx_unpack
 new hash: 0dbd26e398fb21e58e7a555aed4556c4683e630f48bb21370ce461d2bfbc31b5
 score=0.9922864489998511 (hidden), label=1.0
create_new_entry
 new hash: 05dd53abe60bff3c46b3b644109a369668a55f2ac35d472dc9ddac594f44c601
 score=0.9922864489998511 (hidden), label=1.0
break_optional_header_checksum
 new hash: 9ac72d4670de48e3ddb8225d7b22e6962c4eaa81496cfe67af2ad1e632435eda
 score=0.9922864489998511 (hidden), label=1.0
imports_append
 new hash: 43aeb6bea8a59f99feca4e55494ec40cb8c6244fe172fd051dbe7c854b35420d
 score=0.9913104099202542 (hidden), label=1.0
```

```
 new sha256: 035a21ec0cbe8484026e377cf2a3d062dd26d1d5b41ce358f7feb44685030bed
 remove_debug
 new hash: 035a21ec0cbe8484026e377cf2a3d062dd26d1d5b41ce358f7feb44685030bed
 score=0.9827242556530589 (hidden), label=1.0
 create_new_entry
 new hash: 57edc74a4dd5df7d32a9b531d34da9cb9ddc8a315026a95881735ab3c85702c9
 score=0.8973482610331932 (hidden), label=0.0
 episode is over: reward = 10.0!
```

Evasion Results

- Agent training: 15 hours for 100K trials (~10K games x 10 turns ea.)
- Using malware samples from VirusShare

Evasion rate on 200 holdout samples



Cross-evasion: detection rate on VirusTotal (average)

- from 35/62 (original)
- to 25/62 (evade)

Summary

- Gym-malware: https://github.com/endgameinc/gym-malware
  - No knowledge of target model needed
  - Manipulates raw binaries: no malware source / disassembly needed
  - Produces variants that preserve **format** and **function** of original
  - Make it better!

- Machine learning models quite effective, even under attack

- All machine learning models have blind spots

Thank you!

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